

# **Project Charter**

## **Client**

### **Regional Hospital System**

A multi-hospital healthcare organization seeking to improve quality of care, reduce avoidable readmissions, and strengthen operational performance.

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## **Engagement**

### **Hospital Readmission Reduction Analytics**

An end-to-end healthcare analytics consulting case study designed to identify factors associated with 30-day hospital readmissions and recommend data-driven interventions that improve patient outcomes and reduce unnecessary healthcare utilization.

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## **Project Background**

Hospital leadership has observed an increase in 30-day readmissions across several service lines. Elevated readmission rates negatively affect patient outcomes, increase operational burden, and may result in reduced reimbursement under the Centers for Medicare & Medicaid Services (CMS) Hospital Readmissions Reduction Program (HRRP).

Leadership has commissioned the Healthcare Analytics Team to investigate the underlying drivers of readmissions and identify opportunities for targeted intervention.

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## **Business Goal**

Reduce avoidable 30-day hospital readmissions through evidence-based clinical and operational improvements supported by healthcare analytics.

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## Primary Stakeholders

| Stakeholder                    | Primary Interest                                                   |
|--------------------------------|--------------------------------------------------------------------|
| Chief Executive Officer (CEO)  | Financial sustainability, quality performance, strategic oversight |
| Chief Quality Officer (CQO)    | Readmission reduction, quality improvement, patient outcomes       |
| Chief Medical Officer (CMO)    | Clinical quality, evidence-based care, physician engagement        |
| Director of Care Management    | Care coordination, discharge planning, transitional care           |
| Hospital Operations Leadership | Resource utilization, bed capacity, operational efficiency         |

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## Consulting Team

Healthcare Analytics Consulting Team

Project Leader: Wai Fun Leong

Role: Provide data-driven insights that support executive decision-making through data integration, statistical analysis, predictive modeling, visualization, and actionable recommendations.

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## Scope of Work

This engagement includes:

- Business problem definition
- Data quality assessment
- SQL-based ETL and data integration
- Exploratory data analysis
- Statistical analysis
- Predictive modeling for readmission risk

- Executive dashboard development
- Strategic recommendations

## **Out of Scope**

The following activities are outside the scope of this engagement:

- Primary data collection (e.g., patient surveys, clinical observations, or prospective data collection).
  - Electronic Health Record (EHR) system implementation or integration.
  - Deployment of a production-ready clinical decision support system or mobile application.
  - Real-time patient risk prediction in a live clinical environment.
  - Clinical validation or implementation of predictive models for patient care.
  - Regulatory approval, compliance auditing, or changes to hospital policies and procedures.
  - Post-implementation monitoring and measurement of intervention outcomes.
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## **Assumptions**

This engagement assumes that:

- The provided datasets accurately represent hospital operations and patient encounters.
  - Patient data have been de-identified to protect privacy.
  - Historical data are sufficient to evaluate readmission patterns.
  - Stakeholders have agreed on the definition of a 30-day readmission.
  - The analytical findings are intended to support decision-making and should complement, not replace, clinical judgment.
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## **Expected Deliverables**

- Executive summary
- Data quality assessment report
- SQL ETL workflow
- Exploratory data analysis
- Predictive readmission model
- Tableau executive dashboard

- Business recommendations
  - GitHub documentation
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## Success Criteria

The engagement will be considered successful if it:

- Identifies key demographic, clinical, and operational factors associated with 30-day readmissions.
  - Provides clear, data-driven insights that support executive and clinical decision-making.
  - Develops a predictive risk stratification model capable of identifying patients at elevated risk for readmission.
  - Delivers an interactive executive dashboard for monitoring readmission trends and key performance indicators.
  - Presents practical, evidence-based recommendations to support quality improvement and reduce avoidable readmissions.
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## Project Timeline

| Phase  | Activity                                   |
|--------|--------------------------------------------|
| Week 1 | Business discovery & project planning      |
| Week 2 | Data audit & SQL ETL                       |
| Week 3 | Exploratory data analysis                  |
| Week 4 | Statistical analysis & predictive modeling |
| Week 5 | Dashboard development                      |
| Week 6 | Executive recommendations & presentation   |